



TEAM NAME: FIGHT CLUB

COLLEGE NAME: Sri Venkateswara College of Engineering

PROBLEM STATEMENT: Tackling the environmental and logistical impact of mega-events by automating food, room, and parking management for zero-waste wedding functions

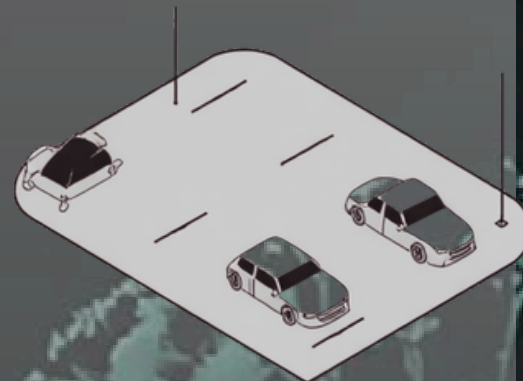
PROBLEM STATEMENT

The absence of real-time guest data in traditional weddings creates a data-blind planning environment, forcing organizers to over-procure resources based on unverified estimates rather than actual, precise, and measurable demand.

Tackling Indian wedding wastage, our AI-driven system optimizes food, rooms, and parking through real-time data, converting resource-heavy mega-events into sustainable, zero-waste celebrations via smart, data-driven logistical management.



- **Who is Affected?** Families, guests, and the local ecosystem burdened by financial loss, logistical chaos, and environmental waste
- **Why it is Important?** Vital for transforming mega-events into sustainable, zero-waste celebrations through data-driven resource optimization.
- **Existing Challenges?** Relying on guest "guesswork" leads to massive food over-provisioning and "blind" parking/room mismanagement.



INDIA TODAY

The big fat Indian wedding is quietly triggering a climate chaos

In India, much of the wedding-related food waste is due to oversized buffets, inaccurate guest estimates, and a general lack of awareness about food donation.

- **The Resource Surplus Paradox:** Massive environmental and economic waste in Indian weddings caused by unoptimized food and accommodation surpluses.
- **Logistical Gridlock:** Inefficient tracking of parking and navigation causes guest congestion and venue mismanagement, disrupting event sustainability and flow.

SOLUTION

Approach

- **QR-Digital Bridge:** Employs personalized QR codes on invitations to establish a seamless digital connection between guests and organizers.
- **Granular Data Capture:** Aggregates real-time attendance, meal preferences, and logistical requirements (parking/rooms) well in advance.
- **Predictive Resource Mapping:** Replaces traditional "guesswork" with precise, data-driven requirements for catering and accommodation.
- **Centralized Command Hub:** Provides a real-time dashboard for organizers to monitor guest arrivals and resource consumption dynamically



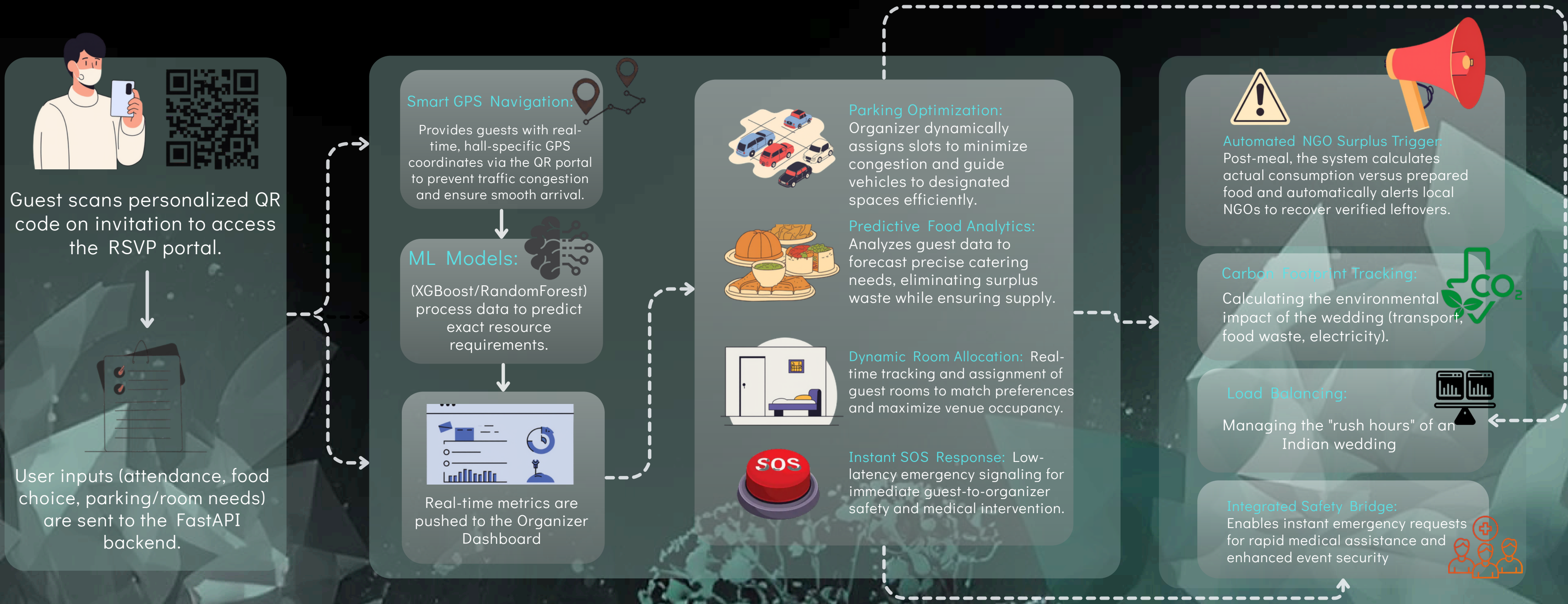
Key Innovation

- **Precision Resource Allocation:** By leveraging real-time guest data for exact headcounts and room occupancy, we eliminate the traditional 20% food "safety buffer" and ensure 100% utilization of accommodation facilities.
- **Dynamic Logistical Optimization:** The system integrates live parking monitoring with GPS-guided navigation to prevent venue gridlock, ensuring a frictionless guest experience through intelligent, data-driven traffic and hall management.

- **Impact Analytics & Certification:** Quantifies carbon and waste savings through a "Sustainability Score," awarding "Zero-Waste Certifications" to incentivize eco-friendly planning.
- **AI Logistics & Redistribution:** Dynamically manages parking and rooms while triggering automated "Surplus Alerts" for immediate NGO food redistribution.



FLOW DIAGRAM



TECHNICAL REQUIREMENTS

Languages & Frameworks

Frontend:



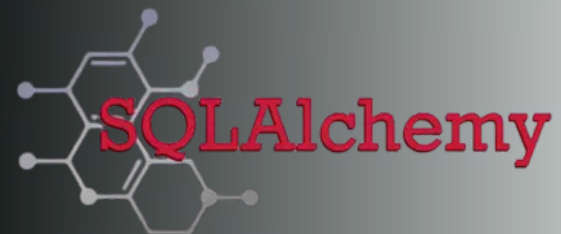
ML Layer:



Backend:



Data Modeling:



Styling:



Tools

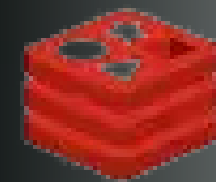
PostgreSQL



Database



Security



redis

Caching



supabase

Backend Services



External
APIs



docker



DevOps

git

NOVELTY / USE CASE

USE CASE

- **Smart Wedding Management:** Optimizes food, parking, and accommodation for large weddings using real-time guest data.
- **Zero-Waste Event Planning:** Predicts exact resource requirements to minimize food waste and environmental impact.
- **Large Event Crowd Management:** Improves parking allocation, guest navigation, and check-ins to avoid congestion.
- **NGO Food Recovery System:** Automatically connects surplus food from events to nearby NGOs for redistribution.
- **Sustainable Event Certification:** Provides analytics on carbon footprint and waste reduction for eco-friendly celebrations.

- **QR-Driven Guest Interaction:** Personalized QR invitations allow guests to instantly access RSVP, check-in, event details digitally.
- **AI-Powered Resource Forecasting:** Machine learning predicts exact food, room, and parking requirements, reducing event waste.
- **Real-Time Organizer Dashboard:** Organizers monitor guest check-ins, RSVP status, and resource usage through a live control panel.
- **Automated Food Redistribution System:** The platform detects surplus food and triggers alerts to nearby NGOs for immediate recovery.



BUSINESS MODEL / TARGET MARKET

BUSINESS MODEL



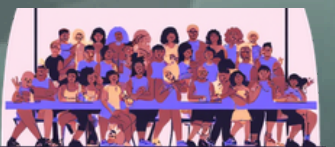
- **Event Planner Subscription:** Organizers subscribe to manage food, parking, and guest logistics efficiently.
- **Venue Integration Service:** Marriage halls integrate the system for guest flow, parking, and room management.
- **Premium Sustainability Reports:** Provides food savings analytics, carbon tracking, and Zero-Waste certification.
- **NGO Food Redistribution Network:** Connects event venues with NGOs to redistribute surplus food.



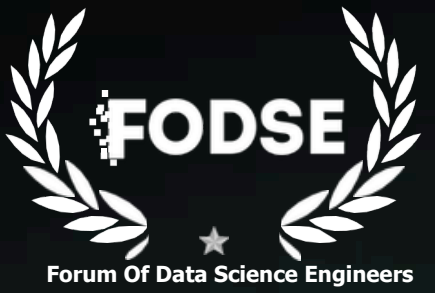
TARGET MARKET



- **Wedding Event Organizers:** Planners managing large weddings who need efficient guest and resource coordination.
- **Marriage Halls & Convention Centers:** Venues requiring smart systems for guest flow, parking, and room management.
- **Catering Service Providers:** Caterers using demand prediction to reduce over-preparation and food waste.
- **Large Events & Gatherings:** Festivals, corporate events, and conventions with unpredictable attendance.
- **NGOs & Food Recovery Networks:** Organizations that collect and redistribute surplus food from events.



TEAM DETAILS



ROLE	NAME	STREAM/DEPT	YEAR	COLLEGE
LEADER	SANJAY V	AIDS	III	SVCE
MEMBER	SUDESHWAR T S	AIDS	III	SVCE
MEMBER	SANJAY K	AIDS	III	SVCE
MEMBER	SIVAKARTHICK K	AIDS	III	SVCE
MEMBER	VISHWA A	AIDS	III	SVCE
MEMBER	RAMASOCKALINGAM M	AIDS	III	SVCE